

Team Contribution & Responsibilities

Project: Inventory or High-Value Asset Risk Oracle: A Hardware-Anchored Assessment System

Team Members: Sanjeev Jha, Jasreen Bagga

Project Phase	Task / Responsibility	Contributor(s)
1. Research & Ideation	Project Ideation & Conceptualisation: Defining the core "Oracle Problem" and designing the physical-to-digital bridge for Real-World Assets (RWAs).	Jasreen Bagga, Sanjeev Jha
	Literature Review: Researching traditional finance vulnerabilities (e.g., Qingdao Metals Scandal) and reviewing existing smart contract literature.	Sanjeev Jha, Jasreen Bagga
	Slogan Strategy & Value Proposition: Developing the core pitch ("Blueprint for Trustless Custody").	Sanjeev Jha, Jasreen Bagga
2. Core Development	Basic Code Architecture: Drafting the initial logic for the Python edge environment and the foundational structure of the Solidity smart contract.	Sanjeev Jha, Jasreen Bagga
	Hardware Assembly & IoT Calibration: Wiring the Raspberry Pi 4, HX711 ADC, and load cell; engineering the signal processing algorithms (weight.py, test_weight.py).	Sanjeev Jha, Jasreen Bagga

	Advanced Middleware & Cryptography: Developing the asynchronous polling loop, sensor-triggered camera integration, and SHA-256 JSON proof generation (App.py, camera.py).	Sanjeev Jha
3. Web3 Integration	Smart Contract Finalisation: Writing the deterministic on-chain LTV (Loan-to-Value) mathematics and state transition logic (Contract.txt).	Sanjeev Jha
	RPC Integration & Security: Connecting the hardware edge device to the Sepolia Testnet using Alchemy and implementing cryptographic transaction signing (chain.py).	Sanjeev Jha
4. UI/UX & Testing	Front-End Dashboard Development: Building the real-time Flask UI, asynchronous JavaScript polling, dynamic LTV health bars, and immutable audit trail (index.html).	Sanjeev Jha, Jasreen Bagga
	Threat Modeling & QA: Testing edge cases (environmental vibrations, sensor creep) and ensuring the local predictive state accurately mirrors the on-chain consensus.	Sanjeev Jha, Jasreen Bagga
5. Deliverables	Poster Presentation Design: Translating the complex 9-Step Risk Oracle Cycle and technical architecture into a high-impact visual format.	Jasreen Bagga, Sanjeev Jha,

	Drafting the Final Report: Compiling the multi-chapter technical engineering report, formatting the codebase, and writing the system methodology.	Sanjeev Jha, Jasreen Bagga
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Digital Signature:



Sanjeev Jha



Jasreen Bagga

